HUNG PHAM

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EDUCATION

George Mason University Fairfax, VA
Bachelor of Science in Computer Science *August 2021 – May 2025*
Relevant Courses: Natural Language Processing, Data Mining, Operating System, Computer Vision
Awards: Mason Distinction Scholarship ($68,000), Tech Talent Award ($8,000)

TECHNICAL SKILLS

Languages: Python, Java, C, C#, SQL, JavaScript, HTML/CSS
Frameworks: PyTorch, TensorFlow, keras, pandas, NumPy, OpenCV, Matplotlib, scikit-learn, Spring Boot
Developer Tools: Git, Docker, make, Unity, Robot Operating System (ROS), MySQL
Operating System: Ubuntu, Kali Linux, Windows
Spoken Languages: Vietnamese, English

EXPERIENCE

Security Research Assistant September 2024 – Present
Security of Emerging Computer System And Technologies Lab (SECSAT) *Fairfax, VA*

- Spearheaded an original first-author publication (in progress) on large language models (LLMs) security.
- Investigated a novel attack using Python that targeted the KV cache in LLMs to extract system prompt
- Experimented with pruning methods on LLMs, revealing trade-offs between compression and vulnerability.
- Reviewed machine learning system security literature to refine research with latest insights.

Robotics Research Assistant August 2024 – Present
Robotic Anticipatory Intelligence And Learning Group (RAIL) *Fairfax, VA*

- Engineered ML-based robotic navigation using Python and ROS to enhance LoCoBot indoor path planning.
- Reduced model size by 74% and lowered computation in a PyTorch neural network while preserving accuracy.
- Deployed an optimized model on LoCoBot, cutting navigation cost by 21% versus Dijkstra greedy planner.
- Configured LoCoBot hardware, integrating LiDAR and panoramic camera into the planner for navigation.

Backend Developer Intern June 2022 – Aug 2022
FPT Telecom *Hanoi, VN*

- Designed and developed a robust back-end application using Spring Boot with the help of a mentor.
- Optimized a MySQL database, improving retrieval efficiency.
- Gained hands-on experience with API development and implementation through integrating REST API.

Co-founder & Vice President August 2023 – Present
Mason Pool Club *Fairfax, VA*

- Established the GMU Pool Club, recruiting 50+ members in one semester.
- Led the club to secure 3rd place in a Virginia regional tournament.

PROJECTS

Pyquaticus | *Python, Pygame, NumPy, Docker, Git* August 2024 – Present

- Transitioned 2D graphics to 3D in Pyquaticus, a maritime CTF simulation made by Navy Research Lab.
- Utilized Python and Pygame to develop and test 3D rendering features.
- Worked in a team with a mentor from NRL to develop functionality, optimizing code for enhanced performance.

Bag Of Visual Words Classifier | *Python, OpenCV, Numpy, Tensorflow* August 2023 – December 2023

- Developed a custom Bag of Visual Words model from scratch for image classification.
- Achieved classification accuracy of 95% on the Fashion MNIST dataset.
- Implemented an improved SIFT with bilateral filters, K-nearest neighbor to enhance model robustness.
- Validated the custom model against a pre-built model and a CNN, achieving superior performance.